BRYAN BERMUDEZ

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Software Engineering student (B.S., May 2027) who builds end to end: custom ESP32 firmware for a Bluetooth smart watch, Linux automation for a self-hosted home server, and an original 2D sandbox game. Seeking a new-grad software engineering role starting May 2027. U.S. citizen, authorized to work without sponsorship.

EDUCATION

Florida Institute of Technology — Melbourne, FL | B.S. Software Engineering · GPA 3.14 · Expected May 2027
Relevant coursework: Operating Systems, Computer Graphics, Software Engineering, Software Testing, Intro to Project Management, Differential Equations & Linear Algebra

TECHNICAL SKILLS

- **Languages:** Java, Python, C/C++, SQL, Bash, Assembly (basic)
- **Embedded & hardware:** ESP32, Arduino, microcontroller firmware, Bluetooth, touchscreen peripherals, power/battery optimization, rapid prototyping
- **Systems & tools:** Linux (Ubuntu), shell scripting, Git & GitHub, Docker, Godot, Jira/Scrum, SolidWorks CAD
- **Software practice:** object-oriented design, data structures, relational databases, LLM output evaluation & QA, debugging on real hardware

PROJECTS

Bluetooth Smart Watch — ESP32, C/C++ · Jun 2025 – Sep 2025
- Designed and built a wrist-sized wearable from scratch: selected components, laid out the board, and wrote the firmware driving a touchscreen, Bluetooth radio, and battery on an ESP32.
- Diagnosed three failures in the first prototype (fast battery drain, dropped Bluetooth link, unreliable touch input) by measuring power draw and logging connection events; rewrote the connection-handling and power-management code for prototype two.
- Validated the second prototype through multi-day wear testing until it held up in daily use, reaching up to 4 days of battery life per charge.

2D Sandbox Game — Godot · solo indie project, in development
- Designing and programming an original block-based sandbox game from scratch: tile/chunk world system, procedural world generation, player physics, and inventory.
- Built a custom 2D lighting engine and an occlusion manager to keep rendering cost down as the world grows.
- Designed a modular, data-driven spell system so new spells are composed from reusable components without touching core code.
- Created all pixel-art tiles, blocks, and character sprites; planning a commercial release.

Home Network Solutions (HNS) — Bash, Ubuntu Server, Docker · CSE 4101 team project (4 people) · Fall 2026, in progress
- Building a self-hosted home server platform; own the Bash installers that download, install, and configure the platform's tools on Ubuntu.
- Built the password hashing system for user credentials.
- Collaborate through Git with teammates who own the Jellyfin and Pi-hole services; services run in Docker containers.

Selected Course Projects — github.com/Bryan1Bermudez
- **Operating Systems (CSE 4001, C/C++):** implemented new system calls in the OS/161 teaching kernel; wrote programs using the process API (fork/exec/wait), inter-process communication, and multi-thread synchronization with semaphores.
- **Computer Graphics (CSE 5280, Python):** built a ray tracer, a space-carving 3D reconstruction, a robot-arm kinematics solver, camera-trajectory animation with Lie-group interpolation, and a multi-particle simulation.
- **ECPLA Marketplace (Intro to Project Management, CSE 4410, in progress):** team-built marketplace web app developed in Scrum sprints tracked in Jira, with sprint status reporting. **Software Testing (CSE 4425):** dynamic black-box test design.

EXPERIENCE

Handshake AI — AI Model Trainer (contract, remote) · Aug 2025 – Sep 2026
- Evaluated and graded large-language-model outputs against detailed rubrics for correctness, reasoning, and instruction-following to produce training and validation data.
- Reviewed ~100 model responses per week across coding, math, language, and visual-pattern tasks.
- Completed a full internal/client project end to end; earned the Model Validation 2 – Expert credential (Credly).

ProPlay Games — Inventory Data & Pricing Associate · Miami, FL · Jul 2025 – Aug 2025
- Priced trading-card inventory against live market data and recorded price, ID, and condition for each card in the store database.
- Maintained accurate records for high-value inventory and stocked display binders, processing hundreds of cards per week.

CERTIFICATIONS & LANGUAGES

- **Model Validation 2 – Expert**, Handshake AI (Credly) · **SolidProfessor Technical Certification** (SolidWorks CAD), May 2021
- Fluent in English and Spanish